In [2]:
```
import pandas as pd
import numpy as np

df = pd.read_csv("C:/Users/mestr/Downloads/archive (8)/emails.csv")

print(df.head())
print(df.shape)
```

```
  Email No.  the  to  ect  and  for  of    a  you  hou  ...  connevey  jay  \
0   Email 1    0   0    1    0    0   0    2    0    0  ...         0    0
1   Email 2    8  13   24    6    6   2  102    1   27  ...         0    0
2   Email 3    0   0    1    0    0   0    8    0    0  ...         0    0
3   Email 4    0   5   22    0    5   1   51    2   10  ...         0    0
4   Email 5    7   6   17    1    5   2   57    0    9  ...         0    0

   valued  lay  infrastructure  military  allowing  ff  dry  Prediction
0       0    0               0         0         0   0    0           0
1       0    0               0         0         0   1    0           0
2       0    0               0         0         0   0    0           0
3       0    0               0         0         0   0    0           0
4       0    0               0         0         0   1    0           0

[5 rows x 3002 columns]
(5172, 3002)
```

In [3]:
```
print(df.columns)
print(df.isnull().sum())
```

```
Index(['Email No.', 'the', 'to', 'ect', 'and', 'for', 'of', 'a', 'you', 'hou',
       ...
       'connevey', 'jay', 'valued', 'lay', 'infrastructure', 'military',
       'allowing', 'ff', 'dry', 'Prediction'],
      dtype='object', length=3002)
Email No.     0
the           0
to            0
ect           0
and           0
             ..
military      0
allowing      0
ff            0
dry           0
Prediction    0
Length: 3002, dtype: int64
```

In [4]:
```
print(df['Prediction'].value_counts())
```

```
Prediction
0    3672
1    1500
Name: count, dtype: int64
```

In [5]:
```
X = df.drop(['Email No.', 'Prediction'], axis=1)
y = df['Prediction']
```

In [6]:
```
print(X.shape)
print(y.shape)
```

```
(5172, 3000)
(5172,)
```

In [7]:
```python
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [8]:
```python
print(X_train.shape)
print(X_test.shape)
```

```
(4137, 3000)
(1035, 3000)
```

In [9]:
```python
from sklearn.neighbors import KNeighborsClassifier

knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)

y_pred_knn = knn.predict(X_test)
```

In [10]:
```python
from sklearn.metrics import accuracy_score

print("KNN Accuracy:", accuracy_score(y_test, y_pred_knn))
```

```
KNN Accuracy: 0.8628019323671497
```

In [11]:
```python
from sklearn.svm import SVC

svm = SVC()
svm.fit(X_train, y_train)

y_pred_svm = svm.predict(X_test)
```

In [12]:
```python
print("SVM Accuracy:", accuracy_score(y_test, y_pred_svm))
```

```
SVM Accuracy: 0.8173913043478261
```

In [13]:
```python
print("KNN Accuracy:", accuracy_score(y_test, y_pred_knn))
print("SVM Accuracy:", accuracy_score(y_test, y_pred_svm))
```

```
KNN Accuracy: 0.8628019323671497
SVM Accuracy: 0.8173913043478261
```

In [ ]:
```python

```